DIGITAL PAYMENTS FRAUD DETECTION

An Evidence-Based Benchmark of Class-Weighted Logistic Regression and Random Forest

RabTech Task 6 — Data Science Research Capstone & Whitepaper

Research question
How do a class-weighted Logistic Regression model and a class-weighted Random Forest compare in detecting rare fraudulent transactions when the decision threshold is selected using a validation set?

Dataset: European card transaction benchmark (284,807 transactions; 492 fraud cases).

Analysis: chronological train/validation/test split; threshold selection on validation data; final evaluation on held-out test data.

Prepared from the recorded notebook outputs supplied for this project. Performance values below are the experiment's actual saved metrics; no live banking data were used.

Abstract

Payment fraud detection is a rare-event classification problem: fraudulent cases represent only a very small share of transactions, so accuracy alone can conceal poor detection. This study benchmarks class-weighted Logistic Regression and Random Forest on a public, anonymized European card-transaction dataset containing 284,807 transactions, including 492 fraud cases (0.1727%). A chronological 60/20/20-style split produced 170,884 training rows, 56,961 validation rows and 56,962 test rows. A decision threshold was selected on validation data and then applied to the held-out test set. Random Forest achieved test precision 0.8906, recall 0.7600, F1 0.8201, precision-recall average precision 0.8193 and ROC-AUC 0.9490. It identified 57 fraud cases, missed 18 and generated 7 false positives. Logistic Regression achieved higher precision (0.9259) and ROC-AUC (0.9820) but lower recall (0.6667), F1 (0.7752) and average precision (0.7437), with 50 true positives, 25 false negatives and 4 false positives. On the recorded experiment's selected operating thresholds, Random Forest offered the stronger precision-recall balance and average precision, while Logistic Regression generated fewer alerts but missed more fraud. These results support evaluating models using threshold-aware metrics and operational costs rather than accuracy alone. The benchmark does not establish production readiness: the data are historical, anonymized and not a live digital-wallet or real-time bank feed.

Keywords: payment fraud, class imbalance, Random Forest, Logistic Regression, precision-recall, threshold selection, risk triage.

1. Introduction

Digital payment systems need to identify suspicious transactions quickly while limiting unnecessary declines, manual reviews and customer friction. Fraud is rare relative to legitimate activity. A model that labels every transaction as legitimate can appear highly accurate while detecting no fraud at all. This makes the choice of evaluation metrics central to any useful fraud-detection study.

The task is not simply to maximize a generic score. A fraud team must balance missed fraud (false negatives) against alerts that are not fraudulent (false positives). The acceptable balance depends on transaction value, review capacity, customer impact and the cost of fraud. Model probabilities also need an operating threshold: the default 0.5 threshold is not necessarily appropriate for an imbalanced dataset.

This capstone conducts a reproducible benchmark on a public anonymized transaction dataset. It compares a linear baseline with a tree-based ensemble, uses a validation set to select a decision threshold, and evaluates both models on a separate test set. The analysis is designed as an educational research benchmark, not as evidence that a model is ready to block live payments.

2. Research Question and Objectives

Primary research question. How do class-weighted Logistic Regression and class-weighted Random Forest compare on held-out fraud detection when the decision threshold is selected on validation data?

The study has four objectives:

- Measure the scale of class imbalance and document the chronological data split.
- Compare the models using precision, recall, F1, average precision / PR-AUC and ROC-AUC.
- Inspect confusion-matrix counts to make the operational trade-off tangible.
- Translate findings into practical recommendations and identify limitations before any real-world deployment.

3. Literature Background

Prior work consistently treats payment-card fraud detection as a challenging imbalanced-learning problem. The dataset used here is a widely reused public benchmark assembled from European cardholder transactions. Its anonymized transformed features make it useful for comparing methods, but they also limit interpretation of individual feature meanings and do not provide the complete context available to a bank's production fraud platform.

Research on feature extraction and sampling has examined how imbalance treatments and representation choices affect fraud classification. Salekshahrezaee, Leevy and Khoshgoftaar (2023) study the effects of feature extraction and data sampling on credit-card fraud detection. This motivates careful reporting of preprocessing and split design rather than treating model choice as the only determinant of performance.

A systematic review published in 2025 specifically examines machine learning for credit-card fraud detection under original class imbalance, underscoring that evaluation under the natural class distribution is an important issue. In this capstone, the held-out test results are reported without resampling the test set, so the reported confusion counts reflect the highly imbalanced benchmark test subset.

This project compares two common approaches. Logistic Regression is a comparatively simple linear baseline that can be easier to inspect and calibrate, while Random Forest combines many decision trees and can represent nonlinear interactions. Neither algorithm is universally best: performance depends on data, features, costs, thresholding and operational constraints.

4. Dataset and Data Understanding

The dataset contains 284,807 transactions and 492 fraud labels. The measured fraud rate is 0.1727486%, meaning fewer than two transactions per thousand are labelled fraudulent. Features include anonymized transformed variables commonly named V1 through V28, along with Time and Amount, and the target label Class. The transformed variables are not directly interpretable as ordinary customer or device attributes.

Dataset property	Recorded value
Total transactions	284,807
Fraud cases	492
Fraud rate	0.1727486%
Training partition	170,884 rows
Validation partition	56,961 rows
Test partition	56,962 rows

The class imbalance means that overall accuracy would be an inadequate headline measure. Precision answers, “Of the transactions flagged, how many were fraud?” Recall answers, “Of the fraud cases, how many did the model catch?” F1 is the harmonic mean of precision and recall. Average precision summarizes the precision-recall ranking across thresholds, while ROC-AUC measures ranking separation across true-positive and false-positive rates. For rare events, precision-recall measures are particularly useful because they reflect the positive class more directly.

5. Methodology

5.1 Experimental design

The notebook uses a chronological partition of the data into training, validation and test subsets. The recorded sizes are 170,884, 56,961 and 56,962 rows respectively. Chronological splitting is intended to better approximate the challenge of evaluating on later transactions than a fully random split would, although it does not eliminate all forms of temporal or population shift.

5.2 Preprocessing

The notebook standardizes the Time and Amount variables and passes the remaining transformed variables through. The transformation is fit as part of the model pipeline using training data, which helps avoid using held-out test information to estimate scaling parameters. The label is Class. The public benchmark's anonymized features limit feature-level business interpretation.

5.3 Models

Two supervised classifiers were evaluated: (1) class-weighted Logistic Regression, used as a linear baseline; and (2) Random Forest, using class-weight handling and an ensemble of decision trees. Class weighting makes the minority class more influential during training without artificially changing the held-out test distribution. The comparison is not exhaustive; boosting, calibrated models and anomaly-detection methods remain candidates for later experiments.

5.4 Threshold selection and evaluation

For each model, a decision threshold was selected using validation data rather than the final test set. The selected threshold was then applied to the untouched test partition. This separation matters because selecting a threshold on the test set would make the reported test metrics optimistic. The experiment summary records the chosen thresholds as 0.2663 for Random Forest and approximately 0.99998 for Logistic Regression. The latter is unusually high and should be interpreted carefully: it reflects the notebook's validation-based F1 selection procedure and model score distribution, not a generally recommended production threshold.

5.5 Metrics

Metric	Interpretation
Precision	TP / (TP + FP): fraction of alerts that are fraud.
Recall	TP / (TP + FN): fraction of fraud cases detected.
F1-score	Harmonic mean of precision and recall.
Average precision (PR-AUC summary)	Summary of precision-recall ranking across thresholds.
ROC-AUC	Ranking discrimination across false-positive and true-positive rates.
Confusion counts	TP, FP, FN and TN counts at the selected threshold.

6. Results

The saved experiment summary identifies Random Forest as the model with the higher test average precision. Its precision was 0.8906 and recall was 0.7600, giving an F1-score of 0.8201. Logistic Regression had higher precision (0.9259) but lower recall (0.6667), and its average precision was lower. The comparison is summarized below.

Test metric	Random Forest	Logistic Regression
Selected validation threshold	0.266317	0.999977
Precision	0.890625	0.925926
Recall	0.760000	0.666667
F1-score	0.820144	0.775194
Average precision (PR-AUC)	0.819318	0.743704
ROC-AUC	0.948998	0.982016

6.1 Confusion-matrix interpretation

Test count	Random Forest	Logistic Regression
True positives (fraud caught)	57	50
False positives (non-fraud flagged)	7	4
False negatives (fraud missed)	18	25
True negatives (non-fraud correctly cleared)	56,880	56,883

At the selected thresholds, Random Forest detected 7 more fraud cases than Logistic Regression (57 versus 50) while producing 3 more false positives (7 versus 4). It missed 18 fraud cases compared with 25 for Logistic Regression. In this test set, there were 75 fraud cases implied by the confusion counts (57 + 18 and 50 + 25), so Random Forest recall was 57/75 = 76%, while Logistic Regression recall was 50/75 = 66.7%. These counts make the trade-off clearer than a single score.

6.2 Interpreting ROC-AUC versus average precision

Logistic Regression produced the higher ROC-AUC (0.9820 versus 0.9490), yet Random Forest produced the higher average precision (0.8193 versus 0.7437) and better F1 at the selected operating points. These metrics summarize different aspects of ranking quality. In a rare-event setting, a strong ROC-AUC does not automatically imply better precision-recall performance at the region of interest. Model selection should therefore be tied to the operational objective and alert budget.

6.3 Threshold caveat

The Logistic Regression threshold reported by the experiment is approximately 0.99998. This value is close to one and merits review before the result is treated as a stable operational setting. It may arise because the validation threshold maximized F1 among candidate scores under the fitted model's probability distribution. A follow-up should plot validation precision and recall across thresholds, confirm threshold-selection code and evaluate threshold stability over multiple chronological windows. The test metrics remain the outputs of the recorded experiment, but a threshold should not be copied into production without further validation.

7. Discussion

The recorded experiment favors Random Forest when the primary objective is to catch more fraud while retaining high precision. At its selected threshold it detected 76% of test fraud cases, with 89.1% precision and only seven false-positive alerts in the test partition. Logistic Regression was more selective, with 92.6% precision and four false positives, but it caught a smaller share of fraud and had lower F1 and average precision.

This does not mean Random Forest is always the best model. In a real payment system, a false negative can lead to financial loss, while a false positive can cause a declined legitimate transaction or a manual-review burden. The relative costs are institution-specific. A bank might choose a higher-recall threshold for high-value transactions, or route uncertain cases to step-up verification rather than immediately decline them. This benchmark has no customer-impact or transaction-cost labels, so it cannot calculate the financial value of either strategy.

The results also show why accuracy should not be used as the sole success measure. The majority class dominates the dataset. Reporting precision, recall, F1, average precision and confusion counts makes it easier to understand both fraud capture and the burden of false alerts. The held-out chronological test is a stronger evaluation design than reporting training performance alone, but a single benchmark and single split still provide limited evidence of robustness.

8. Business Implications and Recommendations

8.1 Human-in-the-loop risk triage

A practical fraud workflow should not rely on one model score as an automatic final decision in every case. A risk-triage design can assign high-risk transactions to manual review or step-up verification, allow low-risk transactions to proceed subject to existing controls, and reserve an uncertainty band for additional signals. The specific bands must be calibrated to real operating costs and policy constraints, which were not measured in this study.

8.2 Alert capacity and false positives

The test confusion counts show seven false positives for Random Forest and four for Logistic Regression. Those counts are small in the benchmark test set, but they should not be extrapolated directly to a large live payment stream. At production scale, even a low false-positive rate can create substantial review volume. Before launch, teams should estimate alert volume per day, reviewer capacity, false-decline rates, and customer friction using representative, authorized data.

8.3 Monitoring and governance

- Track precision, recall, average precision, alert rate and fraud loss by time period, channel and transaction segment.
- Monitor data drift and concept drift; fraud patterns and legitimate customer behavior change over time.
- Reassess thresholds when review capacity, fraud costs or payment mix changes.
- Keep an auditable record of model version, training window, feature pipeline, threshold and evaluation metrics.
- Provide a human review and appeal path for high-impact transaction decisions.
- Run controlled shadow deployment before automated blocking and compare model decisions with confirmed outcomes.

8.4 Recommended next experiments

- Validate the threshold-selection procedure and plot precision-recall curves with thresholds annotated.
- Repeat evaluation across several chronological windows to estimate metric variability.
- Compare class-weighted Random Forest with gradient boosting and calibrated Logistic Regression.
- Evaluate cost-sensitive metrics using realistic, approved estimates of fraud loss and review cost.
- Test probability calibration and segment-level performance before proposing operational thresholds.

9. Limitations, Ethics and Responsible Use

The data are a public, historical, anonymized benchmark, not a live feed from a bank, UPI provider or digital-wallet company. The project does not access account identifiers, device fingerprints, GPS location or current customer behavior. It therefore does not demonstrate real-time transaction ingestion or device/location verification. Such features should only be used when legally authorized, necessary, secure and transparently governed.

The feature transformation reduces the interpretability of the original variables. The analysis uses one chronological split and does not provide confidence intervals or a formal significance test. The reported threshold values are selected from validation data and may not remain optimal under future data distributions. The benchmark also does not quantify monetary fraud losses, customer friction, or investigation workload. These limitations prevent claims of production effectiveness.

Ethical deployment requires privacy protection, access controls, data minimization, clear governance, monitoring for disparate impact where suitable and lawful, and human oversight for consequential decisions. A fraud score should be treated as a risk signal rather than proof that a person committed fraud. Model decisions should be auditable, and legitimate users need a way to resolve mistaken flags.

10. Conclusion

This capstone compared class-weighted Logistic Regression and Random Forest on a highly imbalanced public transaction dataset using chronological training, validation and test partitions. Random Forest achieved the higher average precision (0.8193) and F1-score (0.8201), and caught 57 of 75 fraud cases in the test set. Logistic Regression had higher precision (0.9259) and ROC-AUC (0.9820), but caught 50 of 75 fraud cases and missed more fraud. The choice between them depends on whether the operating goal prioritizes fraud capture, fewer false alarms, or a cost-sensitive balance.

The central finding is methodological as much as algorithmic: evaluation must reflect class imbalance, thresholds must be chosen without using the test set, and confusion counts should be interpreted alongside ranking metrics. Random Forest is the stronger candidate for further study in this experiment, not a production-ready decision system. Future work should examine threshold stability, temporal robustness, calibrated scores, business costs and authorized operational data.

11. Reproducibility and Artifact Guide

The accompanying project repository should contain the notebook, README, requirements.txt, requirements-lock.txt, whitepaper source, and the small experiment outputs. The raw dataset is approximately 150.8 MB and should not be committed to a public GitHub repository. The README should explain how to obtain the public dataset and place it at data/raw/creditcard.csv, how to install dependencies, and how to run the notebook. Include the actual exported CSV/JSON outputs and keep the report consistent with those outputs.

The current notebook saves results under notebooks/reports/outputs because its output path is relative to the notebook working directory. Copies were placed in the root reports/outputs folder for report generation. To avoid confusion in future runs, the notebook should use one documented output location and create the directory before writing files.

References

[1] Kaggle / Machine Learning Group – ULB. Credit Card Fraud Detection dataset. https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

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[4] Scikit-learn documentation. LogisticRegression, RandomForestClassifier, precision-recall metrics and ROC-AUC. https://scikit-learn.org/stable/

Research integrity note: All numeric performance results in this document are transcribed from the experiment_summary.json output supplied by the project author. Dataset description and literature references are cited above. No real banking data or live production system was used.